

Oceanic Strike-Slip Faults, Transform Boundaries & Fracture Zones

Authoritative Reference: Ben A. van der Pluijm & Stephen Marshak, *Earth Structure* (2nd Ed., Chapters 8, 14 & 19)

Comprehensive Chapter Masterclass: Strike-Slip Faults & Transform Boundaries

Wilson's Revolutionary Concept, Mechanics of Ridge-Ridge Offsets & Oceanic Fracture Zones

1. J. Tuzo Wilson & The Formulation of Transform Faults

Prior to 1965, all strike-slip faults were interpreted as simple transcurrent faults that cut across preexisting geological features. In 1965, the Canadian geophysicist **J. Tuzo Wilson** published a revolutionary paper titled "*A new class of faults and their bearing on continental drift*" (*Nature*), recognizing that oceanic strike-slip faults that offset mid-ocean ridges represent an entirely new mechanical and kinematic category: **transform faults**.

*"In the preferred sense, transform faults are **plate boundaries at which lithosphere is neither created nor destroyed**... A transform fault links two other plate boundaries (divergent or convergent) and accommodates the relative horizontal motion between the separating plates. Seismicity along ridge-ridge transform faults is strictly confined to the active segment connecting the two ridge crests."*

— Van der Pluijm & Marshak, *Earth Structure* (2nd Ed., Table 8.2, p. 171 & Chapter 19, p. 450)

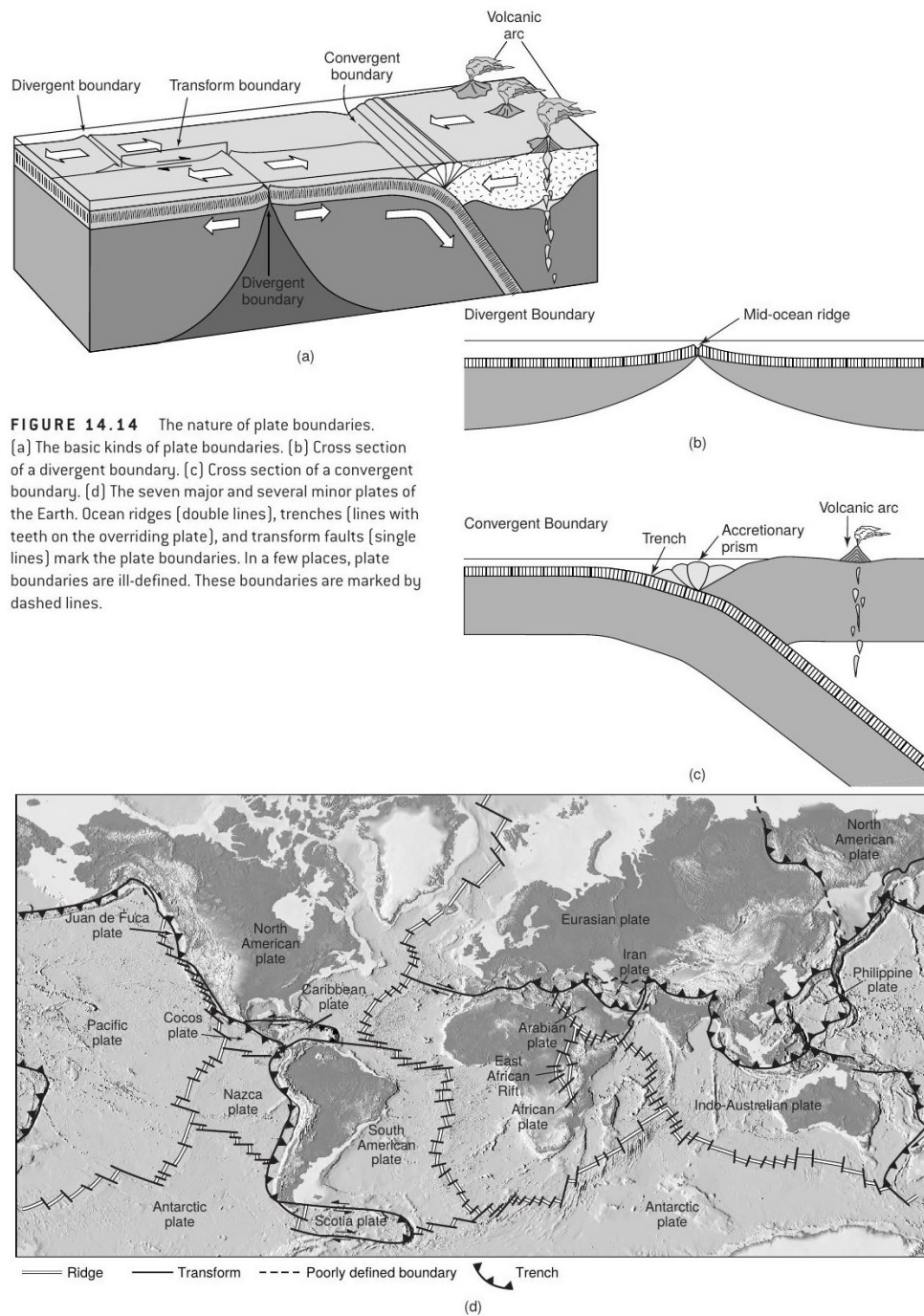


Figure from Van der Pluijm & Marshak (Chapter 19): Kinematic geometry of an oceanic ridge-ridge transform fault. Notice that the relative sense of motion between the two spreading ridges (arrows) is opposite to what simple apparent offset of the ridge axes would suggest for a conventional transcurrent fault.

TABLE 19.1	TERMINOLOGY FOR STRIKE-SLIP FAULTS
Flower structure	An array of faults in a strike-slip fault zone that merges at depth into a near-vertical fault plane, but near the ground surface diverges so as to have shallower dips. In a positive flower structure, there is a component of thrusting on the faults, and in a negative flower structure, there is a component of normal faulting.
Horsetail	A group of fault splays at the endpoint (fault tip) of a fault; the splays diverge to define a fanlike array in map view.
Lateral escape	The strike-slip movement of a crustal block in a direction perpendicular to the regional convergence direction in a collisional orogen; the block is essentially squeezed laterally out of the way of the colliding blocks.
Lateral ramp	A surface connecting two non-coplanar parts of a thrust fault or normal fault (i.e., a ramp that is roughly parallel to the transport direction). Strike-slip movement occurs on a lateral ramp.
Non-coplanar	An adjective used to describe a geometric arrangement of two planes in which the planes have the same attitude, but are not aligned end-to-end to form a single plane.
Pull-apart basin	A sedimentary basin that forms at a releasing bend along a strike-slip fault.
Releasing bend	In the context of describing a strike-slip fault, it is a map-view bend, in the fault plane, whose orientation is such that the block on one side of the fault pulls away from the block on the other, causing transtensional deformation.
Restraining bend	In the context of describing a strike-slip fault, it is a map-view bend, in the fault plane, whose orientation is such that the block on one side of the fault pushes against the block on the other, causing transpressional deformation.
Stepover	A geometry that occurs where the end of one fault trace overlaps the end of another, non-coplanar fault trace.
Strike-slip duplex	An arrangement of sigmoidal-shaped fault splays (in map view) at a stepover, a restraining bend, or a releasing bend, whose geometry in map view resembles the cross-sectional geometry of the duplexes that occur in thrust-fault or normal-fault systems.
Strike-slip fault	Any fault on which displacement vectors are parallel to the strike of the fault, in present-day surface coordinates. The use of this term is purely geometric, and has no genetic, tectonic, or size connotation.
Tear fault	A traditional term for strike-slip faults that occur in a thrust sheet and accommodate differential displacement of one part of a thrust sheet relative to an adjacent part; in more recent jargon, a tear fault is a vertical lateral ramp.
Transcurrent fault	A strike-slip fault that has the following characteristics: it dies out along its length; the displacement across it is less than the length of the fault; the length of the fault increases with time and continued movement; displacement on the fault is greatest at the center of the fault trace and decreases toward the endpoints (tips) of the fault.
Transfer fault	A fault trending roughly perpendicular to the axis of a rift that links together two non-coplanar parts of the rift; transfer faults accommodate map-view offset of the locus of extension. Kinematically, transfer faults are a type of transform fault. If a rift succeeds in evolving into a mid-ocean ridge, the transfer fault becomes an oceanic transform fault.
Transform fault	A strike-slip fault that has the following characteristics: once formed, displacement across it can be constant along the length of the fault; displacement across it can be much greater than the length of the active fault; its length can be constant, increase, or decrease with time; it terminates at another fault. Transform faults can be lithosphere plate boundaries, in which case they terminate at intersections with other plate boundaries.
Transpression	A combination of strike-slip and compressional deformation; transpression occurs where a fault is not parallel to the map projection of regional displacement vectors, so that there is a component of compression across the fault (leading to shortening).
Transtension	A combination of strike-slip and tension; transtension occurs where a fault is not parallel to the map projection of regional displacement vectors, so that there is a component of tension across the fault (leading to extension).
Wrench fault	A traditional synonym for a strike-slip fault. It was commonly used in the petroleum geology literature, typically in reference to a regional-scale continental strike-slip fault. We abandon the term, as it tends to be used inconsistently.

Figure from Van der Pluijm & Marshak (Chapter 19): Six geometric classes of transform faults connecting combinations of spreading ridges (divergent boundaries) and subduction zones (convergent boundaries).

2. Diagnostic Distinctions: Transform Faults vs. Other Strike-Slip Faults

In structural geology (Van der Pluijm & Marshak, Chapter 8, Table 8.2), strike-slip faults are subdivided into distinct genetic classes based on their tectonic scale and boundary terminations:

Strike-Slip Fault Type	Tectonic Nature & Termination	Displacement Distribution	Seismic Activity Pattern
Transform Fault	Major lithospheric plate boundary; terminates abruptly at other plate boundaries (ridges, trenches)	Constant displacement along the active plate boundary segment	Earthquakes strictly confined to segment between ridge crests; fossil extensions (fracture zones) are aseismic
Transcurrent (Wrench) Fault	Intraplate strike-slip fault; dies out into rock within a single plate at tip lines	Displacement is maximum at center and decreases to zero at both tips	Earthquakes distributed along the entire length of the fault plane
Tear Fault	Local strike-slip fault bounding a thrust sheet or graben segment to accommodate differential advance	Limited to the moving thrust nappe	Shallow, local seismicity during thrust emplacement
Scissor Fault	Rotational fault where displacement changes sign across a pivotal axis of zero slip	Variable along strike (rotational hinge)	Local block tilting



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 13

Question Statement

Q.13: In ocean, regional strike-slip faults associated with lithospheric plate boundaries are known as:

(a) Transform faults

(b) Tear faults

(c) Wrench faults

(d) Scissor faults

Correct Answer: (a) Transform faults



Regional strike-slip faults that offset oceanic ridges and function as conservative lithospheric plate boundaries are universally designated as transform faults.

Exhaustive Option-by-Option Critical Analysis

(a) Transform faults

CORRECT

Why it is uniquely correct: In global tectonics and structural geology (Van der Pluijm & Marshak, Table 8.2, p. 171; Wilson, 1965; Kearey, Klepeis & Vine, 2009), **transform faults** are explicitly defined as regional strike-slip faults that constitute primary **conservative plate boundaries** connecting segments of mid-ocean ridges or subduction zones. Across the global ocean floor (such as the Romanche and Charlie-Gibbs Fracture Zones in the Atlantic, and the Mendocino and Clarion Transforms in the Pacific), these faults accommodate horizontal motion between major tectonic plates without generating or destroying lithosphere.

(b) Tear faults

INCORRECT

Why it is incorrect: A *tear fault* is a localized intraplate strike-slip fault that develops within a single thrust sheet or hanging wall block, accommodating differential forward motion or lateral termination of thrust nappes. Tear faults are minor structural features confined to individual fold-thrust belts and are not regional lithospheric plate boundaries.

(c) Wrench faults

INCORRECT

Why it is incorrect: *Wrench fault* (or transcurrent fault) is an older descriptive term for large, high-angle strike-slip faults that occur **within continental plates** (e.g., the Great Glen Fault in Scotland). Crucially, transcurrent/wrench faults terminate at tip lines within intact crust rather than ending at plate boundary junctions, and their slip diminishes toward both ends.

(d) Scissor faults

INCORRECT

Why it is incorrect: A *scissor fault* (or pivotal fault) is a fault characterized by differential rotation around an axis perpendicular to the fault plane, such that displacement increases in opposite directions away from a neutral pivot point. It is not an oceanic plate boundary.



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Plate Tectonic Fault Classification & 8 Exam-Grade Multiple-Choice Questions

Classification of Strike-Slip Faults by Scale and Geological Setting

Classification	Primary Geodynamic Setting	Boundary Terminations	Motion Relative to Spreading	Representative World Examples
Oceanic Transform Fault	Ocean basins, lithospheric plate boundaries	Spreading ridge axes or subduction zones	Constant slip across active segment	Romanche, Vema, Charlie-Gibbs Transforms
Continental Transform Fault	Continental lithospheric plate boundaries	Triple junctions or oceanic ridge/trench systems	Accommodates full relative plate motion	San Andreas Fault (California), Alpine Fault (NZ)
Transcurrent (Wrench) Fault	Intracontinental strike-slip	Dies out into ductile shear zones or tip lines	Variable along strike; zero at tips	Great Glen Fault, Altyn Tagh Fault
Tear Fault	Fold-and-thrust belts, accretionary wedges	Terminates at floor/roof thrusts	Compartmentalization of thrust nappes	Jura Mountains tear faults, Moine thrust tears

8 High-Yield Practice Questions on Transform Faults & Strike-Slip Tectonics

1. Who first proposed the existence and mechanics of transform faults in 1965?

- (a) Alfred Wegener (b) Harry Hess (c) J. Tuzo Wilson (d) Fred Vine

Answer: (c) J. Tuzo Wilson — Wilson published the foundational paper demonstrating that transform faults link divergent and convergent plate boundaries.

2. Along an oceanic ridge-ridge transform fault, where do earthquakes occur?

- (a) Across the entire length of the fracture zone extending across the whole ocean basin
- (b) Exclusively along the active segment between the two offsetting ridge crests
- (c) Only at the exact center of the fracture zone
- (d) Earthquakes never occur on transform faults

Answer: (b) — Between the ridge segments, adjacent lithospheric plates move in opposite directions, generating seismicity; outside this active offset, plates move in the same direction at equal speeds, forming aseismic fracture zones.

3. If a mid-ocean ridge trending north-south is stepped to the right (dextral apparent offset), what is the actual sense of slip along the active transform fault?

- (a) Dextral (right-lateral)
- (b) Sinistral (left-lateral)
- (c) Pure dip-slip normal
- (d) Pure thrust

Answer: (b) Sinistral (left-lateral) — In a ridge-ridge transform, spreading from the two offset ridges forces the block between them to slide in the opposite sense (sinistral) to the apparent geographic offset of the ridge crests.

4. The San Andreas Fault in California is classified tectonically as a:

- (a) Mid-plate intraplate fault
- (b) Continental transform fault connecting the East Pacific Rise with the Mendocino Triple Junction
- (c) Local tear fault in a fold belt
- (d) Subduction megathrust

Answer: (b) — The San Andreas Fault is a classic right-lateral continental transform plate boundary connecting spreading ridges and triple junctions.

5. Oceanic fracture zones extending beyond the active ridge segments represent:

- (a) Active subduction trenches
- (b) Aseismic scars marking juxtaposed lithosphere of different ages and thermal subsidence histories
- (c) Active mid-ocean ridges
- (d) Collisional mountain belts

Answer: (b) — Fracture zones are fossil, inactive traces of transforms where crust of different ages (and therefore different depths) are juxtaposed.

6. Which term describes a regional strike-slip fault that terminates within a single tectonic plate without connecting other plate boundaries?

- (a) Transform fault
- (b) Transcurrent (wrench) fault
- (c) Spreading center
- (d) Benioff zone

Answer: (b) Transcurrent (wrench) fault — Transcurrent faults are intraplate features whose slip dies out toward both tips.

7. In a pull-apart basin formed along a strike-slip fault system, what type of stepover geometry is required in a right-lateral (dextral) fault?

- (a) Left stepover (releasing bend) (b) Right stepover (releasing bend) (c) Planar continuous trace (d) Perpendicular intersection

Answer: (b) Right stepover (releasing bend) — A right step on a right-lateral fault creates local crustal extension and subsidence, forming a rhomboidal pull-apart basin.

8. What is the fundamental boundary condition that distinguishes conservative plate boundaries from convergent and divergent boundaries?

- (a) Lithosphere is created
(b) Lithosphere is consumed
(c) Lithospheric surface area is conserved (neither created nor destroyed)
(d) Rocks are converted to diamond

Answer: (c) — Conservative boundaries (transforms) preserve total surface area of oceanic/continental lithosphere.